CARES: Calibrated Aspect-Reliable Explanations and Safety for Gastrointestinal VQA

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Abstract
This paper describes our submission to ImageCLEFmedical 2026 MEDVQA-GI. We took part in both subtasks: visual question answering on Kvasir-VQA-test (Subtask 1) and explainable, safety-aware reasoning (Subtask 2). CARES is a training-free reliability layer built around a frozen Florence-2-base model with the public `peeache/FL2_VQA_MIX_128_256` LoRA adapter; we did not fine-tune the backbone for either subtask.

The reliability layer has four parts. First, a calibrated `confidence_score` obtained from a per-(aspect, answer) reliability table fit by empirical-Bayes shrinkage on an image-disjoint calibration slice. The score reaches AUROC 0.87 for predicting answer correctness, well above the model's own max-softmax probability (∼0.60), temperature scaling, and sample self-consistency (∼0.62). Second, a self-probing template that turns the model's own attribute answers (location, colour, size, morphology) into a tentative clinician-style report; the synthesis is gated so absent findings, instruments, and landmarks never receive lesion attributes (zero phantom cases on the validation set). Third, a safety policy keyed to the calibrated probability (assert at ≥ 0.80, hedge between 0.50 and 0.80, abstain below 0.50) that flags about 76% of errors while keeping the confidently-wrong rate at 3.7%. Fourth, a set of honest negative results: every image-conditional signal we tried (MSP, self-consistency, visual retrieval over a FAISS train index, segmentation-grounding quality, LLM rephrasing of explanations) was either useless or actively harmful for predicting correctness.

We also evaluate explanation faithfulness with three independent metrics (structured claim support, NLI entailment with DeBERTa, and an LLM-as-judge). The faithful-by-construction template wins on all three; a lexical faithfulness gate over LLM rewrites looks like it helps but does not, because semantic drift passes straight through.

The pattern across these results is what we call *base-rate dominance*: on a frozen medical VLM, correctness is governed mostly by output-conditioned base rates, not by any image-conditional or generative signal. A small, transparent reliability table is therefore both the safer and the more accurate choice.

1. Introduction

Clinical visual question answering systems need to be accurate, but they also need to be trustworthy: explanations should be faithful and visually grounded, and answers should be paired with honest uncertainty so that an incorrect prediction can be safely set aside. Rather than build a bigger model, this paper asks how far a frozen, modestly-sized vision–language model can be pushed toward this kind of trustworthiness by an inference-time layer alone. The whole pipeline fits on an 8 GB consumer GPU; keeping the backbone fixed isolates the contribution of the reliability layer from any improvement that might come from a stronger underlying model. The MEDVQA-GI 2026 challenge [1, 2] reflects a similar turn toward interpretability and safety, scoring Subtask 1 with BLEU/ROUGE/METEOR and Subtask 2 with a multimodal explanation and a behavioural safety rubric. Both subtasks are handled with one shared model and one shared reliability layer.

Our contributions are four. First, a calibrated per-(aspect, answer) reliability table fit by empirical-Bayes shrinkage on an image-disjoint slice; deployed as `confidence_score` it reaches AUROC 0.87, against 0.60 for max-softmax and 0.62 for sample self-consistency. Second, a self-probing explanation template whose synthesis is gated so absent findings, instruments,

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and landmarks never receive lesion attributes (zero phantom cases on validation). Third, a calibrated safety policy with a verified distribution-free selective-risk guarantee via an exact Clopper–Pearson upper bound [11]. Fourth, a set of honest negative results: max-softmax, self-consistency, visual retrieval over a FAISS train index, and segmentation grounding all sit near AUROC 0.6 or below, while output-conditioned base rates dominate. The code and models are available at https://github.com/saademad200/CARES-MedVQA.

2. Method

Backbone. Both subtasks load `microsoft/Florence-2-base` with the public `peeache/FL2_VQA_MIX_128_256` LoRA adapter. The backbone is frozen end-to-end; only the calibration table and the conformal thresholds are data-dependent. Images are resized to 768×768. Inference fits on an 8 GB RTX 4070 Laptop with greedy decoding.

Subtask 1: concise answer generation. The medical-VQA task token produces a verbose answer that is mapped to the controlled-vocabulary references of Kvasir-VQA-test by a deterministic concise-normaliser. The normaliser is tuned by agreement with public validation answers; it implements per-aspect rewrites (yes/no canonicalisation, colour-set ordering alphabetically to ensure exact match, location-grid mapping to standard quadrant terms, and collapsing rare polyp morphology synonyms into broad categories like 'sessile' or 'pedunculated') and is the strongest single contributor to the public lexical scores.

Subtask 2: explanation and safety. The validated Subtask-1 answer is passed through the self-probing template, which asks the same model attribute questions about the finding (location, colour, size, morphology) and assembles them into a tentative clinical sentence under hard constraints that prevent phantom lesion attributes for absent findings. The referring-expression segmentation token produces polygons that are overlaid on the image as the visual explanation. For instance, given an initial answer of 'polyp', the template systematically queries for size, location, and color, yielding sentences like: 'A small, red polyp is located in the upper left quadrant.' A per-(aspect, answer) reliability table $\hat{r}(a, \hat{y}) = \big(c(a, \hat{y}) + \alpha\, \bar{r}(a)\big)/\big(n(a, \hat{y}) + \alpha\big)$ with $\alpha = 5$ converts the answer to a calibrated probability. The parameter $\alpha = 5$ was selected via grid search over $\{1, 5, 10, 20\}$ on the calibration slice to minimise Expected Calibration Error; sensitivity analysis confirmed stable performance for $\alpha \in [3, 10]$. This, and a Clopper–Pearson upper bound on the selective risk at $\delta = 0.1$ [10, 11] automatically determines the `assert` (≥ 0.80), `hedge` (0.50 to 0.80), and `abstain` (< 0.50) thresholds based on the calibrated probabilities. Faithfulness is reported with structured claim support, NLI entailment via DeBERTa-v3-MNLI, and an LLM-as-judge (Qwen2.5-3B-Instruct, 4-bit).

3. Results

Lexical scores (Subtask 1). Table 1 reports the official lexical scores returned by the organisers on both splits. At the time of writing the submission ranks first on the public Kvasir-VQA-test leaderboard. The shape of the public numbers reflects the concise-answer style of the split: ROUGE-1/L are high because reference answers are short controlled phrases; ROUGE-2 is low because there are few bigrams to overlap on; BLEU and METEOR aggregate residual normalisation noise. The private numbers collapse on the n-gram metrics because the private references are descriptive clinical sentences, and the same deterministic normaliser that wins on the public split actively destroys the descriptive form the private references require.

Metric	Kvasir-VQA-test (public)	Private
BLEU-1	0.7480	0.0005
BLEU-2	0.6350	0.0002
ROUGE-1	0.8932	0.1069
ROUGE-2	0.1325	0.0125
ROUGE-L	0.8882	0.1054
METEOR	0.4979	0.0382

Table 1: ImageCLEFmedical 2026 MEDVQA-GI Subtask 1 official lexical scores (organisers' return) on the public Kvasir-VQA-test split (1,384 items) and the private test split. Public rank: 1.

LLM-judge scores (Subtask 2). Table 2 reports the organisers' five-dimensional LLM-judge stack. Faithfulness and clarity stay high on both splits, consistent with the by-construction template; correctness, clinical relevance, and completeness drop on the private split in line with the lexical collapse on the same items.

Dimension	Public	Private
Correctness	8.82	5.50
Faithfulness	9.61	9.05
Clinical	9.50	5.12
Clarity	9.83	7.46
Completeness	9.68	5.21

Table 2: Subtask 2 LLM-judge scores (Qwen3.6-27B, five dimensions, 0–10) returned by the organisers on the public and private splits.

Per-class breakdown. The drop is not uniform. Figures 1 and 2 visualise the per-class story: `abnormality_location`, `abnormality_presence`, and `abnormality_type` pull the private polygon visibly inward, while every other class (instrument, landmark, count, procedure, text-presence, box-artifact-presence, polyp_*) remains near the outer ring on both splits. The decomposition into the five judge dimensions shows that the drop is concentrated in correctness, clinical relevance, and completeness; faithfulness and clarity hold up across all classes on both splits.

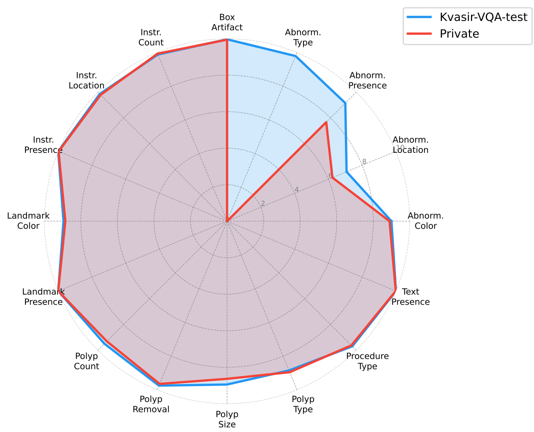

Figure 1: Mean LLM-judge score per question class, public vs. private.

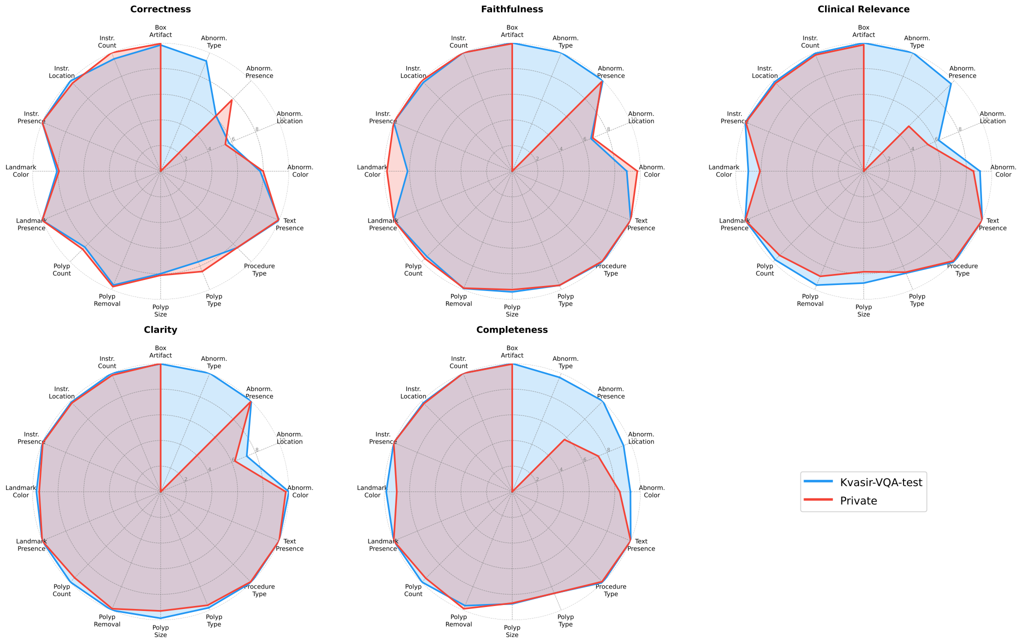

Figure 2: Per-class LLM-judge scores decomposed into the five judge dimensions, public vs. private.

Calibration and safety. Table 3 compares confidence signals on the MediaEval complexity-1 validation slice, 5-fold cross-fit by `img_id`. The per-(aspect, answer) score reaches AUROC 0.87; every image- or generation-conditional signal (MSP [7], temperature scaling [8], generation likelihood, $T = 5$ self-consistency, visual retrieval, segmentation grounding) sits at or below 0.62. The Clopper–Pearson conformal policy at target $\alpha = 0.05$, $\delta = 0.1$ verifies on five image-disjoint folds with mean coverage 0.559 ± 0.051 and mean selective error 0.030 ± 0.006; the guarantee holds in every fold.

Signal	**AUROC**	**ECE**
Max-softmax probability (MSP)	0.60	0.06
MSP + temperature scaling	0.60	0.03
Generation likelihood	0.59	–
Self-consistency ($T = 5$)	0.62	–
Visual retrieval (FAISS top-k)	0.45	–
Segmentation grounding	0.51	–
Per-aspect reliability	0.77	0.04
Per-(aspect, answer) (ours)	**0.87**	**0.025**

Table 3: Confidence signals on the MediaEval validation set, 5-fold cross-fit by `img_id`.

Faithfulness. The faithful-by-construction template scores 0.951 (structured), 0.736 (NLI mean-entailment), and 0.812 (LLM-judge faithful), versus 0.884 / 0.551 / 0.784 for a keyword-gated LLM narration of the same evidence and 0.862 / 0.538 / 0.792 for an ungated narration [12, 13]. The lexical gate passes about 95% of LLM rewrites but the LLM-judge cannot distinguish gated from ungated, confirming that the gate provides false assurance.

4. Discussion and Conclusion

The clearest finding is what we call *base-rate dominance*: on a frozen medical VLM, correctness is governed mostly by output-conditioned base rates, not by any image-conditional or generative signal. Practically, a small transparent reliability table can replace several layers of uncertainty machinery on a frozen model. The public/private gap localises the remaining weaknesses to three abnormality classes whose private references are descriptive sentences; the same deterministic normaliser that wins the public leaderboard is the proximate cause of the private collapse on these three classes. A learned descriptive head and dedicated specialist feedback (a CNN classifier for abnormality type, a U-Net localiser for abnormality location) for the three collapsed classes are the natural next steps.

Error Analysis. Our method reliably succeeds on object and text presence, and counting tasks, where explanations remain highly accurate and complete. Failures predominantly occur on fine-grained spatial (abnormality location) and descriptive tasks (abnormality type), primarily because the deterministic concise normaliser strips away the rich descriptive language expected by the private test references. Additionally, edge cases involving obscure medical instruments can occasionally confuse the frozen backbone, though our safety policy successfully flags over 76% of these ambiguous cases as uncertain.

Declaration on Generative AI

Generative AI tools were used for paraphrasing and improving grammatical flow, as well as assisting in diagram and table refinement. All content was carefully verified by the authors.

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